

L20 — Circular Linked List and Circular Doubly Linked List

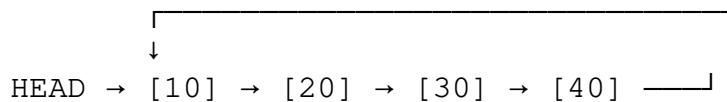
Unit: 2 | Chapter: Linked Lists | BT Level: BT3 | CO: CO2

Learning Objectives

- Explain the structure of circular singly and doubly linked lists
 - Implement insertion, deletion, and traversal for circular linked lists
 - Identify use cases where circular lists are preferred
 - Differentiate circular lists from their linear counterparts
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1. Circular Singly Linked List (CSLL)

In a **circular singly linked list**, the last node's `next` pointer points back to the **HEAD** (not **NULL**).



There is **no NULL** in the chain — the list wraps around.

2. CSLL Operations

2.1 Traversal — must detect when we're back at HEAD

```
def traverse_csll(head):
    if head is None:
        return
    current = head
    while True:
        print(current.data, end=" → ")
        current = current.next
        if current == head:    # Full circle completed
            break
    print("(back to head)")
```

2.2 Insert at Beginning — O(n) (need to update last node)

```
def insert_beginning(self, data):
    new_node = Node(data)
    if self.head is None:
        new_node.next = new_node    # Points to itself
        self.head = new_node
```

```

        return

    # Find last node (the one pointing to head)
    last = self.head
    while last.next != self.head:
        last = last.next

    new_node.next = self.head
    last.next = new_node      # Last node now points to new head
    self.head = new_node

```

2.3 Insert at End — $O(n)$

```

def insert_end(self, data):
    new_node = Node(data)
    if self.head is None:
        new_node.next = new_node
        self.head = new_node
        return

    last = self.head
    while last.next != self.head:
        last = last.next

    last.next = new_node
    new_node.next = self.head    # New node circles back to head

```

2.4 CSLL with Tail Pointer — $O(1)$ insert at end

Using a **tail pointer** (instead of head) avoids $O(n)$ traversal:

```

class CSLL_WithTail:
    def __init__(self):
        self.tail = None    # Points to last node

    def insert_end(self, data):
        new_node = Node(data)
        if self.tail is None:
            new_node.next = new_node
            self.tail = new_node
        else:
            new_node.next = self.tail.next    # New last → old first (head)
            self.tail.next = new_node
            self.tail = new_node

    @property
    def head(self):
        return self.tail.next if self.tail else None

```

With tail pointer: insert at end → $O(1)$, insert at beginning → $O(1)$.

2.5 Delete a Node — $O(n)$

```

def delete_node(self, key):
    if self.head is None:
        return

    last = self.head
    while last.next != self.head:
        last = last.next

    # Deleting head
    if self.head.data == key:
        if self.head == last:      # Only one node
            self.head = None
        else:
            last.next = self.head.next
            self.head = self.head.next
        return

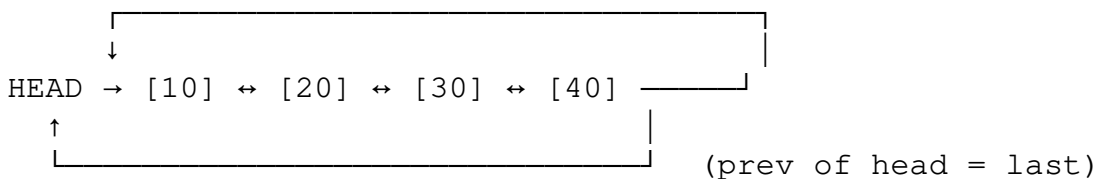
    # Search and delete non-head node
    curr = self.head
    while curr.next != self.head:
        if curr.next.data == key:
            curr.next = curr.next.next
            return
        curr = curr.next

```

3. Circular Doubly Linked List (CDLL)

In a **CDLL**, each node has both `prev` and `next` pointers, and:

- `last.next = first`
- `first.prev = last`



3.1 Node Structure

```

class CDLLNode:
    def __init__(self, data):
        self.data = data
        self.prev = None
        self.next = None

```

3.2 CDLL with Sentinel (Header) Node

The cleanest implementation uses a **sentinel/dummy node**:

```

class CDLL:
    def __init__(self):

```

```

        self.sentinel = CDLLNode(None)    # Dummy node
        self.sentinel.next = self.sentinel
        self.sentinel.prev = self.sentinel

    def insert_after(self, node, data):
        """Insert new node after 'node' - O(1)"""
        new_node = CDLLNode(data)
        new_node.next = node.next
        new_node.prev = node
        node.next.prev = new_node
        node.next = new_node

    def insert_front(self, data):
        self.insert_after(self.sentinel, data)

    def insert_back(self, data):
        self.insert_after(self.sentinel.prev, data)

    def delete_node(self, node):
        """Remove a node - O(1) since we have prev pointer"""
        node.prev.next = node.next
        node.next.prev = node.prev

    def traverse_forward(self):
        curr = self.sentinel.next
        while curr != self.sentinel:
            print(curr.data, end=" ⇔ ")
            curr = curr.next
        print("(sentinel)")

    def traverse_backward(self):
        curr = self.sentinel.prev
        while curr != self.sentinel:
            print(curr.data, end=" ⇔ ")
            curr = curr.prev
        print("(sentinel)")

    def is_empty(self):
        return self.sentinel.next == self.sentinel

```

3.3 Demo

```

cdll = CDLL()
cdll.insert_back(10)
cdll.insert_back(20)
cdll.insert_back(30)
cdll.insert_front(5)
cdll.traverse_forward()    # 5 ⇔ 10 ⇔ 20 ⇔ 30 ⇔ (sentinel)
cdll.traverse_backward()   # 30 ⇔ 20 ⇔ 10 ⇔ 5 ⇔ (sentinel)

```

4. Complexity Comparison

Operation	CSLL (with tail ptr)	CDLL (with sentinel)
Insert front	O(1)	O(1)
Insert back	O(1)	O(1)
Delete given node	O(n)	O(1)
Traverse forward	O(n)	O(n)
Traverse backward	Not possible	O(n)

5. Real-World Applications of Circular Lists

Application	Why Circular?
Round-robin scheduler	Each process gets CPU turn in cycles
Audio/video buffer	Circular buffer (ring buffer) for streaming
Board games (Monopoly)	Players take turns in a cycle
Browser tabs	Ctrl+Tab cycles through tabs
Multiplayer games	Turn rotation among players
Music player	Looping playlist

Ring Buffer (Circular Queue) — O(1) enqueue and dequeue

```
class RingBuffer:
    def __init__(self, capacity):
        self.buffer = [None] * capacity
        self.capacity = capacity
        self.head = 0
        self.tail = 0
        self.size = 0

    def enqueue(self, item):
        if self.size == self.capacity:
            raise Exception("Buffer full")
        self.buffer[self.tail] = item
        self.tail = (self.tail + 1) % self.capacity
        self.size += 1

    def dequeue(self):
        if self.size == 0:
            raise Exception("Buffer empty")
        item = self.buffer[self.head]
        self.head = (self.head + 1) % self.capacity
        self.size -= 1
        return item
```

Summary

- **Circular Singly Linked List:** last.next → HEAD; useful for round-robin applications.
- With a **tail pointer**, both front and back insertions become O(1).

- **Circular Doubly Linked List:** combines CDLL bidirectionality with circular structure.
 - A **sentinel node** simplifies implementation — no NULL checks needed.
 - CDLL with sentinel: $O(1)$ insert/delete anywhere given node reference.
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References

- Cormen et al., *Introduction to Algorithms*, Ch. 10.2 (MIT Press, 2022)
- Skiena, *The Algorithm Design Manual*, Ch. 3.1 (Springer, 2020)